CIRCULAR ELECTRONICS · INNOVATION PITCH

E-WASTE DIGITAL PASSPORT

Giving Every Electronic Device a **Digital Identity**

TRACK | REPAIR | REUSE | REFURBISH | RECYCLE



Digital solution for smarter and traceable e-waste management

02 — THE PROBLEM

E-Waste Has No Digital Identity

Five structural gaps break the chain of information between a device and its final destination.

01

Untracked Devices

Device history is lost after purchase, resale, or ownership transfer.

02

Poor Disposal Decisions

Users don't know whether to repair, reuse, refurbish, or recycle.

03

Fragmented Records

Invoices, warranty, repairs, and part changes are scattered.

04

Lost Material Value

Recoverable components and materials vanish with bad disposal.

05

Limited Recycling Visibility

Collection and recycling centres receive devices blind.

> **"Once a device leaves its first owner, its lifecycle becomes difficult to trace."**

CURRENT LIFECYCLE

BUY → USE → DAMAGE / OBSOLESCENCE → ??? DECISION GAP → E-WASTE



The missing stage is not physical – it is **missing information** at the moment of decision.

03 — OUR SOLUTION

A Digital Passport for Every Device

One scannable identity that carries the device's full record — from factory to recycler.



A tamper-evident **QR / NFC tag** bonded at manufacture.

QR · NFC · CLOUD-LINKED

DEVICE INFORMATION

Model · Manufacturer
Manufacturing date

OWNERSHIP

Purchase date · Warranty
Ownership transfers

DEVICE HEALTH

Battery health · Condition
Component status

SERVICE HISTORY

Repairs · Maintenance
Replaced components

END-OF-LIFE INTELLIGENCE

Reuse potential · Refurbishment potential · Recyclable materials · Disposal recommendation

HOW A USER INTERACTS

01 SCAN QR / NFC
↓
02 OPEN DIGITAL PASSPORT
↓
03 VIEW DEVICE HISTORY
↓
04 GET SMART RECOMMENDATION

AI Recommendation Engine

Reads lifecycle data → suggests the best next action

REPAIR · REUSE · REFURBISH · RESELL · RECYCLE

We don't just identify e-waste — we track the device before it becomes e-waste.

How the Passport Follows a Device

Six actors write to one record. Nothing is created twice, nothing is lost.



WHAT EACH ACTOR WRITES TO THE RECORD

01	MANUFACTURER	Passport issued · specs, materials
02	CONSUMER	Purchase, warranty, usage
03	REPAIR CENTRE	Repairs, replaced parts
04	REFURBISHER	Grade, resale, new owner
05	COLLECTION	Intake scan, routing
06	RECYCLER	Recovered materials, closure

TECHNOLOGY STACK

- QR / NFC
- CLOUD DATABASE
- AI ENGINE
- ANALYTICS

Turning E-Waste into a Circular Economy

Four directions of value created by making a device's history visible.

Environmental

- Reduce improper disposal
- Increase material recovery
- Cut landfill volume
- Embed circular practices

Economic

- Higher resale & refurbishment
- Recover valuable materials
- New business opportunities
- Fewer forced replacements

Social

- Easier responsible disposal
- Greater consumer awareness
- Repair & reuse culture
- Local employment

Technological

- Digital traceability
- QR / NFC identification
- AI-powered recommendations
- Real-time analytics

WITHOUT DIGITAL PASSPORT

BUY → USE → DISCARD

Linear consumption — value ends with the first owner.

WITH DIGITAL PASSPORT

BUY → TRACK → REPAIR → REUSE → REFURBISH → RECYCLE

Circular flow — each stage adds recorded value and feeds the next.

From Linear Consumption → To a Traceable Circular Economy

How Can This Idea Generate Revenue?

A platform monetised across the ecosystem — five recurring streams, one shared record.

01 MANUFACTURER SUBSCRIPTIONS
Pay to create and manage passports for their devices.

02 REPAIR & REFURBISHMENT
Businesses pay for lifecycle records and device management.

03 RECYCLER / COLLECTION
Tracking, categorisation and analytics subscriptions.

04 PREMIUM CONSUMER
Valuation, resale assistance, advanced health reports.

05 DATA & SUSTAINABILITY
Aggregated analytics and sustainability reporting.

MODEL TYPE
B2B SaaS + B2C Premium

Illustrative Year-1 Revenue Scenario

ILLUSTRATIVE MODEL - NOT ACTUAL MARKET REVENUE

Manufacturer subscriptions	20 × ₹50,000 / year	₹10 lakh
Repair / refurbishment partners	100 × ₹2,000 / month	₹24 lakh
Recycling partners	50 × ₹3,000 / month	₹18 lakh
Premium consumer services	optional add-ons	₹5 lakh
Analytics & sustainability reports	institutional	₹5 lakh
TOTAL ILLUSTRATIVE ANNUAL REVENUE		**≈ ₹62 LAKH**

BUSINESS ECOSYSTEM

MANUFACTURERS ₹ →
↓
REPAIR CENTRES ₹ →
↓
REFURBISHERS ₹ →
↓
CONSUMERS ₹ →
↓
COLLECTION CENTRES ₹ →
↓
RECYCLERS ₹ →
↓
E-WASTE DIGITAL PASSPORT PLATFORM

₹ Revenue flows to platform

↻ Value flows to ecosystem

Every Device Has a Story. Give It a Digital Identity.